QUBO Formulation: Quantum Circuit Qubit Routing

Quantum Circuit Qubit Routing

Problem Statement. Given a quantum hardware coupling graph and an initial qubit placement, a set of K candidate routing sequences is pre-generated. Each candidate sequence k incurs a known swap cost c_k, representing the number of two-qubit swaps required, and a nonadjacency penalty p_k, which is zero if the target qubits become adjacent after applying the sequence and a large constant otherwise. The objective is to select exactly one candidate sequence such that the sum of its swap cost and nonadjacency penalty is minimized.

Decision Variables. For each candidate sequence $k \in \{1, \ldots, K\}$, we introduce a binary decision variable $y_k \in \{0, 1\}$. The variable y_k equals 1 if and only if candidate sequence k is selected, and 0 otherwise.

QUBO Derivation. **Step 1 – Objective Function.** The original minimization objective is expressed as

$$\min_y \sum_{k=1}^{K} (c_k + p_k) y_k. \tag{1}$$

Since QUBO solvers minimize energy, we retain this form directly.

Step 2 – Constraint. The requirement to select exactly one sequence is written as

$$\sum_{k=1}^{K} y_k = 1. \tag{2}$$

Step 3 – Penalty Term Expansion. We enforce the constraint by adding a quadratic penalty term with weight $\lambda > 0$:

$$P(y) = \lambda \left(\sum_{k=1}^{K} y_k - 1 \right)^2. \tag{3}$$

Expanding the square yields

$$P(y) = \lambda \left(\sum_{k=1}^{K} y_k^2 - 2 \sum_{1 \le i < j \le K} y_i y_j + 1 \right). \tag{4}$$

Using the idempotent property of binary variables, $y_k^2 = y_k$, we simplify the expression to

$$P(y) = \lambda \sum_{k=1}^{K} y_k - 2\lambda \sum_{1 \leq i < j \leq K} y_i y_j + \lambda. \tag{5}$$

Step 4 – Combined Hamiltonian. Adding the objective and the penalty term gives the full QUBO Hamiltonian $H(y)$:

$$H(y) = \sum_{k=1}^{K} (c_k + p_k) y_k + \lambda \sum_{k=1}^{K} y_k - 2\lambda \sum_{1 \leq i < j \leq K} y_i y_j + \lambda. \tag{6}$$

Grouping linear terms, we obtain

$$H(y) = \sum_{k=1}^{K} (c_k + p_k + \lambda) y_k - 2\lambda \sum_{1 \leq i < j \leq K} y_i y_j + \lambda. \tag{7}$$

Step 5 – Q Matrix and Offset. Reading off the coefficients from the standard upper-triangular QUBO representation $H(y) = \sum_{k=1}^{K} Q_{k,k} y_k + \sum_{1 \leq i < j \leq K} Q_{i,j} y_i y_j + c$, we obtain:

$$Q_{k,k} = c_k + p_k + \lambda, \quad \text{for } k \in \{1, \ldots, K\}, \tag{8}$$

$$Q_{i,j} = -\lambda, \quad \text{for } 1 \leq i < j \leq K, \tag{9}$$

$$c = \lambda. \tag{10}$$

Note that the Hamiltonian contains the term $-2\lambda y_i y_j$ for each pair, which corresponds to $Q_{i,j} = -\lambda$ in the standard QUBO convention.

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any constraint violation increases the total energy above the minimum achievable by a feasible solution. The maximum possible reduction in the objective term $\sum_{k=1}^{K}(c_k+p_k)y_k$ occurs when switching from an infeasible assignment to the best feasible assignment. Since at most one variable can change from 1 to 0 (or vice versa) when moving between feasible and infeasible regions, the maximum objective gain from violating the constraint is bounded by $\max_k(c_k + p_k)$. Therefore, setting

$$\lambda > \max_{k \in \{1,\ldots,K\}} (c_k + p_k) \tag{11}$$

guarantees that the penalty incurred by any infeasible configuration strictly exceeds any potential reduction in the swap cost or penalty terms.

Correctness Argument. Minimizing $H(y)$ over $\{0, 1\}^K$ recovers the optimal routing sequence because (i) all feasible solutions satisfy $\sum_{k=1}^{K} y_k = 1$, causing the penalty term to evaluate to exactly zero, so their energy equals the original objective value plus the constant offset λ; and (ii) any infeasible solution violates the constraint, incurring a penalty of at least λ. By the choice of λ, this penalty strictly dominates any possible decrease in the linear objective terms, ensuring that the global minimum of $H(y)$ must occur at a feasible point. Among feasible points, $H(y)$ reduces to the original objective function, so the minimizer corresponds exactly to the candidate sequence with the minimum total cost.

Illustrative Example. Consider an instance with $K = 3$ candidate sequences. Let the instance parameters be $c_1 = 2, p_1 = 0$; $c_2 = 3, p_2 = 0$; $c_3 = 4, p_3 = 50$. We choose $\lambda = 51$, which satisfies $\lambda > \max\{2, 3, 54\}$. Substituting these values into the general formulas yields the concrete QUBO:

$$
\begin{aligned}
Q_{1,1} &= 2 + 0 + 51 = 53, && (12)\\
Q_{2,2} &= 3 + 0 + 51 = 54, && (13)\\
Q_{3,3} &= 4 + 50 + 51 = 55, && (14)\\
Q_{1,2} = Q_{1,3} = Q_{2,3} &= -51, && (15)\\
c &= 51. && (16)
\end{aligned}
$$

The resulting Hamiltonian is

$$H(y) = 53y_1 + 54y_2 + 55y_3 - 51y_1y_2 - 51y_1y_3 - 51y_2y_3 + 51. \tag{17}$$

Evaluating $H(y)$ over the feasible space $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ gives energies 104, 105, and 106, respectively. Infeasible configurations, such as $(1, 1, 0)$, yield $H(1, 1, 0) = 53 + 54 - 51 + 51 = 107$, which is strictly higher. The global minimum occurs at $y^* = (1, 0, 0)$ with energy 104, correctly selecting candidate sequence 1 as the optimal routing sequence.